

Saber Ayatollahi, CFA, MSc

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SUMMARY

Manager, Quantitative Analytics & Reporting candidate with ~7 years in investment and market risk analytics across Moody's Analytics, EY, and Ortec Finance. Holds delegated sign-off authority on valuation, sensitivity, and risk outputs for multi-asset institutional portfolios of \$5-25bn per engagement. Hands-on with VaR and CVaR optimization, risk decomposition and attribution, stress testing and scenario analysis, derivatives validation, and Python analytics pipelines. CFA charterholder; dual MSc (Financial Modelling, Chemical Engineering).

CORE SKILLS

Investment Risk Analytics (multi-asset) • Value at Risk (VaR) & Conditional VaR (CVaR) • Risk Decomposition & Attribution • Portfolio Exposure & Sensitivity Analysis • Stress Testing & Scenario Analysis • Derivatives Valuation Review (rates, FX, inflation) • Python (pandas, NumPy, SciPy) & SQL • Risk Reporting & Analytics Automation • Independent Model Review & Documentation Standards

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services

Toronto, May 2023 - Present

Risk Analytics Validation & Insight

- Hold delegated sign-off authority on valuation, sensitivity, and risk outputs delivered to multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions before production release.
- Validate derivatives pricing outputs across rates, FX, and inflation, confirming that sensitivities and scenario impacts remain consistent at portfolio-level aggregates.
- Review aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream risk and capital processes.
- Escalate outputs lacking mathematical or economic defensibility; serve as escalation point for client-critical analytical issues with Heads of Risk and investment stakeholders.

Portfolio Exposure, Stress & Scenario Analytics

- Oversee interest-rate exposure and duration analysis under parallel and non-parallel rate shocks across multi-asset institutional portfolios.
- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios across multi-year horizons.

Risk Reporting, Automation & Governance

- Re-engineered manual spreadsheet workflows into scalable Python analytics pipelines with embedded logging, validation, and auditability controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection, cutting development cycle time an estimated 30-40% on comparable modules.
- Prepare analytical summaries for senior stakeholders covering portfolio exposures, scenario impacts, and interest-rate sensitivities, translating quantitative output into committee-ready narratives.
- Serve on the model governance committee, contributing to methodology review, documentation and benchmarking standards, and model-performance assessment for client-delivered analytics.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms on the PFaroe DB (risk & ALM analytics) and PFaroe PM (portfolio management) platforms, acting as technical liaison between client investment teams and product/development functions.

- Migrated all assigned client accounts from a legacy Calypso environment to PFaroe PM, validating model outputs post-deployment.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning reporting requirements with system implementation.
- Coordinated actuarial, finance, IT, and risk stakeholders on milestone delivery, governance documentation, and regulatory-readiness materials.

Ortec Finance Canada

Consultant - ALM & Risk

Toronto, Feb 2019 - Sep 2021

- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, testing allocation robustness through near-optimal frontier analysis.
- Produced risk decomposition and risk attribution analytics (contribution-to-risk, risk budgeting) for multi-asset institutional mandates.
- Built and interpreted stochastic economic scenario generators quantifying interest-rate risk and funding volatility under base-case and stressed environments.
- Delivered ALM studies for Canadian and international pension funds and institutional investors, analyzing funding ratios, duration mismatch, and long-term balance-sheet exposures.
- Conducted cash flow, liquidity, and hedging analysis across interest-rate, inflation, and currency risk, including liability-driven investment (LDI) trade-off analysis.
- Presented study findings and allocation recommendations on-site to investment committees at pension-fund clients.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016